

FFT in Groth16 SNARK Proof Generation

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Overview

In the Groth16 zk-SNARK protocol, Fast Fourier Transform (FFT) is used during the prover's computation to efficiently transform polynomials between coefficient and evaluation forms. This is particularly important in computing the quotient polynomial $h(x)$, which is defined from the relation:

$$\left(\sum_k A_k(x) z_k \right) \left(\sum_k B_k(x) z_k \right) = \left(\sum_k C_k(x) z_k \right) + h(x) Z(x)$$

Here:

- $A_k(x), B_k(x), C_k(x)$ are fixed polynomials derived from the circuit constraints.
- z_k are the witness values (inputs).
- $Z(x)$ is the vanishing polynomial over the evaluation domain.
- $h(x)$ is the quotient polynomial that we wish to compute.

Step-by-Step Procedure

We aim to compute $h(x)$ efficiently using FFT. Starting from the rearranged identity:

$$h(x) = \frac{A(x)B(x) - C(x)}{Z(x)}$$

1. **Choose an Evaluation Domain:** Select a multiplicative subgroup of the field, typically using n -th roots of unity ω^i , where $i = 0, 1, \dots, n-1$.
2. **Evaluate Polynomials:** Use FFT to evaluate the witness-dependent polynomials:

$$A_i = A(\omega^i), \quad B_i = B(\omega^i), \quad C_i = C(\omega^i)$$

This gives their *evaluation form*.

3. **Pointwise Operations:** Perform elementwise (pointwise) operations in the evaluation domain:

$$E_i = A_i \cdot B_i - C_i$$

This gives the evaluation form of $E(x) = A(x)B(x) - C(x)$.

4. **Interpolation to Coefficient Form:** Use inverse FFT to transform E_i back to coefficient form of the polynomial $E(x)$.
5. **Divide by the Vanishing Polynomial:** Compute:

$$h(x) = \frac{E(x)}{Z(x)}$$

where $Z(x) = \prod_{i=0}^{n-1} (x - \omega^i)$. This step is done in coefficient form.

6. **(Optional) Evaluate $h(x)$:** If needed, FFT can be applied again to evaluate $h(x)$ at the domain points.

Small Example over $\mathbb{F}_{17}$

Consider the polynomial:

$$f(x) = 3 + 2x + 5x^2 + x^3$$

Let $\omega = 4 \in \mathbb{F}_{17}$, a primitive 4-th root of unity. Then the evaluation domain is:

$$\{\omega^0, \omega^1, \omega^2, \omega^3\} = \{1, 4, 16, 13\}$$

Evaluate $f(x)$ at these points:

$$f(1) = 3 + 2(1) + 5(1)^2 + (1)^3 = 11$$

$$f(4) = 3 + 2(4) + 5(16) + 64 = 3 + 8 + 80 + 64 = 155 \equiv 2 \pmod{17}$$

$$f(16) = 3 + 2(16) + 5(1) + 16 = 3 + 15 + 5 + 16 = 39 \equiv 5 \pmod{17}$$

$$f(13) = 3 + 2(13) + 5(16) + 16 = 3 + 9 + 12 + 16 = 40 \equiv 6 \pmod{17}$$

Thus, the evaluation form of $f(x)$ using FFT is:

$$[11, 2, 5, 6]$$

Summary

- FFT allows efficient evaluation of polynomials at multiple points in $O(n \log n)$ time.
- In Groth16, we use FFT to compute $A(x)B(x) - C(x)$ in evaluation form.
- We then interpolate back to coefficient form using inverse FFT and divide by the vanishing polynomial $Z(x)$ to obtain $h(x)$.
- FFT is a key optimization that makes proof generation practical for large circuits.